Bilangan Bulat Negatif & Garis Bilangan - Kheyla

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Student

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Topic: Bilangan Bulat Negatif & Garis Bilangan

Status: completed

Lesson Plan

Learning Objectives

- Represent positive and negative integers on a horizontal number line.
- Compare and order negative integers based on their distance from zero.
- Calculate the absolute distance between two integers on a number line.
- Model real-world scenarios, such as temperature changes, using negative numbers.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Kheyla! Before we dive into our new adventure, let's flex our brain muscles with a quick flashback. Check the virtual whiteboard! For Topic 1 (Place Value), can you tell me: (1) In the number 4,520, which digit is in the thousands place? (2) Represent 1,304 using virtual Base-10 blocks—how many 'flats' of 100 do we need? Now for Topic 2 (Diagnostic Fun): (3) If I have a secret number between 50 and 60 that ends in 7, what is it? (4) Quickly sort these numbers from smallest to largest: 450, 405, 540.

Activities:

- *Interactive place value drag-and-drop*
- *Quick-fire number sorting in the chatbox*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Great job, Kheyla! Today, we are going to explore numbers that are 'below zero.' Think about a scientist in a freezer lab studying ice crystals. If the temperature is 5 degrees and drops 10 degrees, where does it end up? We call these 'Negative Integers.' We use them for freezing temperatures, debts, or even going down to the basement in an elevator!

Activities:

• *Viewing a thermometer graphic moving below zero*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Look at this straight line. On the right side, we have our familiar numbers: 1, 2, 3... we call these Positive Integers. Zero is our middle ground. It's neither positive nor negative. What do you think happens if we go to the left of zero?

Activities:

• *Drawing a standard number line 0-10*

4. Problem Introduction (10 minutes)

Teacher Script:

Now, let's reveal the full Number Line! To the left of zero, we have -1, -2, -3... notice they are mirror images. I want you to look at -5 and -2. On a number line, 'greater' always means 'further to the right.' Which one is further to the right? Yes, -2! So, -2 is actually greater than -5. It’s like being less 'in debt' or 'less cold.'

Activities:

• *Marking points on a full -10 to 10 number line*

5. Guided Practice (15 minutes)

Teacher Script:

Let's play 'The Distance Jump.' If a frog starts at -3 and jumps to 2, how many steps did it take? Let's count together: -2, -1, 0, 1, 2. That's 5 steps! The distance is 5. Now, you try: Draw a dot at -4 and a dot at 3 on your screen. What is the distance between them?

Activities:

• *Collaborative number line jumping*

• *Plotting specific temperatures on a scale*

6. Independent Practice (15 minutes)

Teacher Script:

You're a pro! Now, I've sent you a 'Sorting Challenge' in the chat. 1. Sort these: -8, 2, 0, -3 from smallest to largest. 2. A scientist sees the temp is -10°C. It rises 4 degrees. What is the new temp? Type your answers in the chat, then draw them on the whiteboard number line to check your work.

Activities:

- *Chat-based sorting*
- *Word problem solving*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Amazing work today, Kheyla! Remember, with negative numbers, the 'bigger' the negative digit looks (like -100), the 'smaller' its value actually is because it's so far left. Before we log off, please remember to join https://learn.algonova.id/login and head to your lesson dashboard to see your gold stars! See you next time!

Activities:

- *Recap of key findings*
- *Link reminder*

Homework

Medium Questions:

1. Place the following numbers on a number line: -5, -2, 3, -1, 0.
2. Which number is greater, -7 or -3? Draw a number line to prove it.
3. Order these temperatures from coldest to warmest: 4°C, -2°C, -10°C, 0°C.
4. What is the distance between -4 and 1 on a number line?

Hard Questions:

1. A submarine is 50 meters below sea level (-50). It rises 20 meters. What is its new position?
2. Find the sum of the distance from -5 to 0 and the distance from 0 to 3.
3. If you are at -8 and you move 10 units to the right, where do you land?
4. Complete the inequality: -15 [< / >] -12 and explain why.
5. A freezer was at -18°C. A power outage caused it to rise 5 degrees. What is the temperature now?
6. Identify 3 integers that are greater than -4 but less than 2.

Answer Key:

Medium 1: Order on line should be -5, -2, -1, 0, 3.
Medium 2: -3 is greater (it is to the right of -7).
Medium 3: -10°C, -2°C, 0°C, 4°C.
Medium 4: 5 units.
Hard 1: -30 meters.
Hard 2: 5 + 3 = 8.
Hard 3: 2.
Hard 4: < (less than) because -15 is further left than -12.
Hard 5: -13°C.
Hard 6: Any three of: -3, -2, -1, 0, 1.
REVIEW (Place Value) M1: 5 (hundreds).
REVIEW (Place Value) M2: 8,000 + 400 + 0 + 2.
REVIEW (Place Value) H1: 0.
REVIEW (Place Value) H2: 5,630.
REVIEW (Diagnostic) M1: 22.
REVIEW (Diagnostic) M2: 345, 354, 435, 453, 534, 543.
REVIEW (Diagnostic) H1: 54.
REVIEW (Diagnostic) H2: 78.

Parent Reminder

'Dþ Exploring the World of Negative Numbers!

Hi! Today Kheyla explored the 'frozen zone' of math: Negative Integers. She learned how to use a number line to compare cold temperatures and underwater depths. She's doing great—ask her which is 'bigger': -2 or -10!